

Task 8: Mastering INNER, LEFT, RIGHT & FULL JOINS

Tools:

- MySQL Workbench
- Alternatives: PostgreSQL, BigQuery Sandbox

Hints / Mini Guide:

1. Use previously created employees and departments tables.
2. Write an INNER JOIN to fetch employees with their department names.
3. Execute a LEFT JOIN to identify employees without departments.
4. Perform a RIGHT JOIN to list all departments including empty ones.
5. Simulate FULL JOIN using UNION since MySQL does not support it directly.
6. Compare row outputs from each join type.
7. Identify business use cases for each join.
8. Practice optimizing join readability using table aliases.

Deliverables:

- SQL file demonstrating all join types with outputs

Final Outcome:

- Intern can combine data across multiple tables efficiently.

Interview Questions Related To Above Task:

- Difference between INNER and LEFT JOIN?
- When do we use FULL JOIN?
- Why does MySQL not support FULL JOIN?
- What happens if join condition is wrong?
- Real-world use case of LEFT JOIN?

📌 Task Submission Guidelines

- 🕒 **Time Window:**

You can complete the task anytime between 10:00 AM to 10:00 PM on the given day. Submission link closes at 10:00 PM

- 🔍 **Self-Research Allowed:**

You are free to explore, Google, or refer to tutorials to understand concepts and complete the task effectively.

- 🔧 **Debug Yourself:**

Try to resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

- 💰 **No Paid Tools:**

If the task involves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.

- 📁 **GitHub Submission:**

Create a new GitHub repository for each task.

Add everything you used for the task — code, datasets, screenshots (if any), and a short README.md explaining what you did.

- 📌 **Submit Here:**

After completing the task, paste your GitHub repo link and submit it using the link below:

- 👉 [[Submission Link](#)]

Best
of
Luck

